

Is Meshing Always Necessary? The Limited Impact of Rendering Fidelity on VR Spatial Perception

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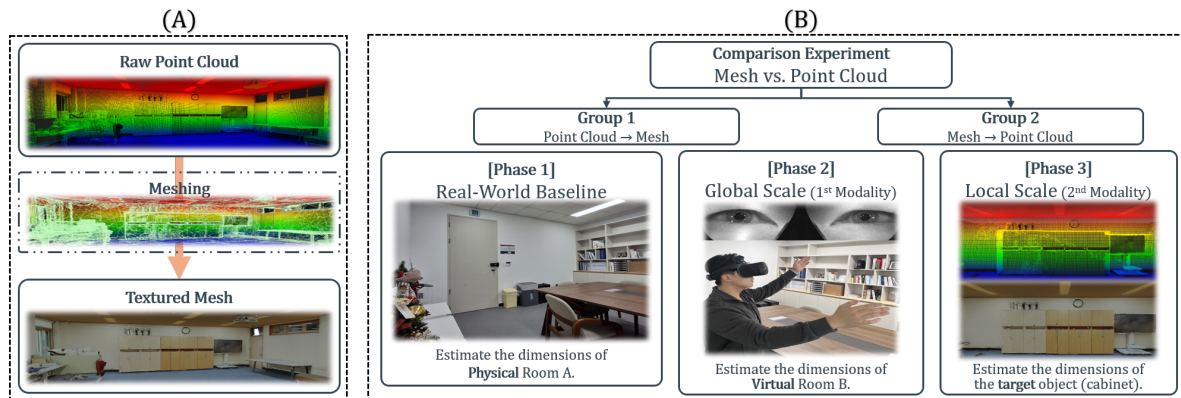


Figure 1: (A) Point-to-surface reconstruction pipeline and (B) comparative experimental design.

ABSTRACT

While textured meshes are widely used in virtual reality, their computational cost raises questions about their perceptual benefit. We conducted an experiment ($N = 15$) utilizing a real-world baseline to normalize individual spatial biases. Results demonstrated comparable global spatial accuracy between representations ($p = 0.921$), confirming the non-inferiority of point clouds. However, local judgments revealed a significant overestimation bias in meshes (+9.69%), consistent with the effects of high visual density, whereas point clouds remained unbiased. Overall, these findings suggest that higher visual fidelity does not inherently improve spatial judgment, highlighting point clouds as an efficient alternative for immersive environments.

Index Terms: Virtual Reality, Spatial Perception, Rendering Fidelity, Point Cloud, Human Factors

1 INTRODUCTION

With advances in LiDAR and photogrammetry, virtualizing the physical world has become increasingly accessible. However, selecting an appropriate representation mode (e.g., point clouds or meshes) remains a critical dilemma between computational efficiency and visual fidelity. While textured meshes are generally favored for their aesthetic superiority [1, 10], they also require extra effort to generate a continuous surface. Moreover, numerous optimization frameworks have been proposed to mitigate the transmission overhead of mesh-based representations in real-time systems [2]. Despite these efforts, the conversion overhead remains a critical bottleneck for real-time applications, particularly when contrasted with the immediate availability of raw point clouds [6].

Beyond computational efficiency, geometric representation choices shape how users perceive and interact with virtual environments, particularly in terms of spatial judgment. While distance

underestimation is common in VR [5], recent studies still explore how various cues can influence perception [7]. A common assumption is that the higher visual quality of mesh automatically leads to better objective spatial understanding. However, we challenge this "Fidelity Myth." Thompson et al. [9] demonstrated that the quality of computer graphics does not significantly affect the accuracy of distance judgments, and Langbehn et al. [3] found that even significant visual blur does not impair distance estimation. Furthermore, recent findings on the Oppel-Kundt illusion indicate that high visual density can actually be counterproductive. Surkys [8] showed that "filled" intervals trigger perceptual overestimation, whereas introducing gaps significantly mitigates this distortion. Consequently, this study questions the necessity of the costly conversion to mesh by comparing the ability of point clouds and mesh to support spatial perception.

In addition, it is crucial to acknowledge that spatial perception is also influenced by intrinsic human factors. Prior work has shown spatial awareness varies across individuals, influenced by factors such as gender [4], and real-world perceptual anchoring [11]. Accordingly, we accounted for these individual variances when comparing point clouds and mesh representations and additionally examine potential perceptual biases induced by rendering style.

2 METHODOLOGY

The experiment comprised a physical room (Room A) used for real-world baseline and a virtual office environment (Room B) used for VR tasks. The VR environment was implemented in Unity and presented using a Varjo VR headset. A within-subjects design was employed, with 15 participants (7 M, 8 F) counterbalanced into two groups (G1: PC→Mesh, G2: Mesh→PC) to mitigate order effects.

The procedure consisted of three phases. In Phase 1 (Real-world Baseline), participants estimated the dimensions ($W \times D \times H$) of Room A to assess individual spatial perception abilities. In Phase 2 (Global Scale), participants viewed Room B in their first assigned modality and estimated its overall spatial dimensions. In Phase 3 (Local Scale), participants experienced the alternative modality and estimated the dimensions of a specific local object (a cabinet), with the task design chosen to mitigate potential carryover effects from the first modality.

Spatial perception was assessed using three complementary metrics, each serving a distinct analytical purpose. First, Absolute

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Percentage Error (APE) was used to quantify overall estimation accuracy, capturing the magnitude of error regardless of direction ($|Estimated - Actual| / Actual \times 100$). Second, Signed Error was computed to identify systematic perceptual bias, indicating whether participants tended to overestimate (positive values) or underestimate (negative values) spatial dimensions. This metric was particularly used to examine directional distortions in local object perception. Third, to account for individual differences in spatial ability, VR performance was normalized using a real-world baseline. Specifically, each participant's VR APE was adjusted by subtracting their corresponding real-world APE, isolating the effect of representation modality from inherent spatial bias. Statistical comparisons between representation conditions were conducted using paired t-tests, while gender-related differences in normalized VR performance were examined using independent t-tests.

3 RESULT

3.1 Global Spatial Perception

Global spatial perception was first evaluated to determine whether representation modality affected overall room-scale accuracy after accounting for individual baseline differences. No significant difference was observed between the two conditions ($p = 0.921$). The mean normalized APE was 6.74% for point clouds and 7.28% for mesh representations. This comparable performance met the criterion for non-inferiority, indicating that point clouds convey global spatial structure as accurately as meshes, despite avoiding computationally expensive surface reconstruction.

3.2 Local Spatial Perception

At the local scale, representation modality produced a clear divergence in spatial judgment. Signed error analysis revealed a strong directional bias for mesh representations, which led to systematic overestimation of object size (+9.69%). In contrast, point clouds showed near-zero bias (−0.54%), indicating perceptually balanced judgments. When accuracy was evaluated relative to individual real-world baselines, point clouds yielded a significant reduction in error (−10.40%), whereas meshes resulted in increased error (+2.60%). The significant contrast between conditions suggests that the high visual density of meshes induces an expansion illusion, reducing precision in fine-grained size estimation.

3.3 Gender Differences

In the real-world baseline, male and female participants exhibited comparable estimation errors (16.13% vs. 19.11%; $p = 0.573$), indicating similar inherent spatial ability. However, after normalization in the VR environment, a significant divergence emerged. Male participants showed near-zero residual error (0.14%), reflecting effective adaptation to virtual cues, whereas female participants retained higher normalized error (12.99%), suggesting differences in recalibrating spatial perception in VR.

Table 1: Comparison of Spatial Perception Metrics between Point Cloud and Mesh Representations.

Metric	Point Cloud	Mesh	p-value
Global Accuracy (Normalized APE)	6.74%	7.28%	0.921
Local Bias (Signed Error)	−0.54%	+9.69%	< 0.001*
Local Accuracy (Performance Change)	−10.40%	+2.60%	< 0.001*

*Significant difference ($p < 0.05$).

Table 2: Analysis of Gender Differences in Real-world Baseline and VR Adaptation.

Context	Metric	Male	Female	p-value
Real World	Baseline APE	16.13%	19.11%	0.573
VR Env.	Normalized APE	0.14%	12.99%	0.006*

*Significant difference ($p < 0.05$).

4 CONCLUSION & DISCUSSION

Our findings challenge the assumption that high visual fidelity guarantees spatial accuracy. While point clouds demonstrated global accuracy comparable to meshes, significant disparities emerged at the local scale. The dense surface of meshes triggered pronounced overestimation, aligning with the Oppel-Kundt illusion, whereas the sparse nature of point clouds mitigated this distortion, offering superior precision for fine-grained judgments.

Furthermore, the considerable gender-based divergence in VR adaptation underscores the variability in cognitive processing of 3D cues among individuals, thereby validating the necessity of our real-world baseline normalization. Ultimately, we provide empirical evidence that justifies prioritizing raw point clouds in real-time and immersive applications. By demonstrating that high-cost meshing is not essential for, and may even degrade, spatial judgment, we explore the potential of point clouds as a perceptually robust and computationally efficient alternative for latency-sensitive environments.

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